

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Mechanical) (Credit System) Degree Examinations

January 2023

19MEE75 - MAINTENANCE AND RELIABILITY ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Whether a maintenance department uses preventive maintenance successfully? Explain.	10	L*2	1	1
	b) List the fundamental problems with the spare parts management system.	10	L4	2	1
2.	a) Describe the benefits of computers in the execution of maintenance tasks.	10	L2	1	1
	b) Illustrate the principles of maintenance for improving the equipment / system availability.	10	L2	2	1
3.	a) List down the different types of maintenance functions and explain any in detail with the help of block diagrams.	10	L2	1	1
	b) How maintenance quality can be improved via maintenance control?	10	L2	2	1
Unit – II					
4.	a) Describe the reliability-centered maintenance methodology.	10	L2	3	1
	b) How is equipment reliability implemented? Describe using crushing equipment as an illustration.	10	L3	4	1
5.	a) How to apply the concepts of severity, occurrence, and detection in failure mode effect analysis? Explain.	10	L4	3	1
	b) Enumerate the reliability evaluation model's broad assumptions.	10	L4	4	1
6.	a) Describe the steps involved in performing failure mode and effect analysis (FMEA).	10	L2	3	1
	b) Mention the different methods applied to increase reliability during the design phase.	10	L2	4	1
Unit – III					
7.	a) What elements are utilized for the increase in reliability?	8	L1	5	1
	b) What part does parallel configuration play in system reliability?	8	L2	5	1
	c) List any two justifications for not evaluating a company's reliability.	4	L2	5	1
8.	a) Describe redundancy. Describe just ONE of its sorts.	8	L2	5	1
	b) What is maintainability? What components of availability are qualitative?	8	L2	5	1
	c) Analyze the data needed for reliability analysis and summarize it.	4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
